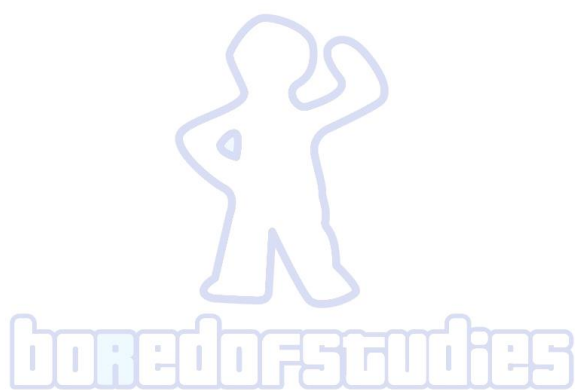




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2023

BORED OF STUDIES TRIAL EXAMINATION

3rd October

Mathematics Extension 2

**General
instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using a black or blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

**Total marks:
100****Section I – 10 marks** (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5–16)

- Attempt Questions 11–16
- Allow about 2 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 A sphere with equation $(x - a)^2 + (y - 2a)^2 + (z - 2a)^2 = 81$ passes through the origin.

Which of the following is a possible value of a ?

- (A) $a = 0$
- (B) $a = 3$
- (C) $a = 6$
- (D) $a = 9$

- 2 Which is the remainder when $x^3 - ix^2 + (1 + i)x - 2 - i$ is divided by $x^2 + 1$?

- (A) -3
- (B) -1
- (C) $ix - 2$
- (D) $2ix - 1$

- 3 Let $\frac{dy}{dx} = \frac{1}{ax^2 - 2abx + c}$ be a differential equation for non-zero constants a, b and c .

Which of the following restrictions ensures that the general solution to the differential equation is $y = A \tan^{-1}(Bx + C) + D$ for some non-zero constants A, B, C and D ?

- (A) $a < 0$, all real $b \neq 0$ and $c < 0$
- (B) $a > 0$, all real $b \neq 0$ and $c < 0$
- (C) $a > 0$, $b > 0$ and all real $c \neq 0$
- (D) $a < 0$, $b > 0$ and all real $c \neq 0$

- 4 At time t seconds, a particle is at the position vector $\underline{r} = 5\underline{i} + (3 - t)\underline{j} - 6\sqrt{t}\underline{k}$.

Which of the below vectors describes the particle's direction of motion at $t = 9$ seconds?

- (A) $-\underline{j} - \underline{k}$ (B) $5\underline{i} - 6\underline{j} - 18\underline{k}$
 (C) $-3\underline{j} - \underline{k}$ (D) $5\underline{i} - \underline{j} - \underline{k}$

- 5 Let x, y and z be real numbers where $x < y$. Suppose that x, y and z satisfy the inequality

$$|x - y| < |x - z| + |z - y|.$$

Which of the following statements is correct?

- (A) The inequality is true for all real z .
 (B) The inequality is only true if $x < z < y$.
 (C) The inequality is only true if $z < x$ or $z > y$.
 (D) The inequality is only true for $z = x$ and $z = y$.

- 6 Let z and w be complex numbers such that $\frac{\pi}{2} < \text{Arg}(z) - \text{Arg}(w) < \pi$.

On the complex plane, the origin and the points representing the complex numbers w, z and $z + w$ are the vertices of a rhombus. Which of the following is correct?

- (A) $\frac{z - w}{z + w} = ki$ where $0 < k < 1$ (B) $\frac{z - w}{z + w} = ki$ where $k > 1$
 (C) $\frac{z + w}{z - w} = ki$ where $0 < k < 1$ (D) $\frac{z + w}{z - w} = ki$ where $k > 1$

- 7 Suppose that a and b are non-zero constants. Which of the following could *possibly* be a unit vector?

- (A) $\frac{1}{a}\underline{i} + b\underline{j} + \underline{k}$ (B) $a^2b^2\underline{i} + 2ab\underline{j} + 2\underline{k}$
 (C) $a\underline{i} + \frac{1}{b}\underline{j} + \frac{1}{a}\underline{k}$ (D) $a^2\underline{i} + 2ab\underline{j} + 2b^2\underline{k}$

- 8 A spring is vertically fixed to the ground. A light object is pushed on top of the spring such that it first compresses the spring downward and is then released such that the object ordinarily performs simple harmonic motion with amplitude A and period ω . Let the centre of this simple harmonic motion be the origin and upwards as the positive direction. The forces acting on the object are gravity and an opposing constant normal force N .

However, on this occasion the object is pushed excessively onto the spring such that when released it loses contact with the spring at its maximum vertical extension.

Let g be the acceleration due to gravity. What is the value of A ?

- (A) $\frac{g}{\omega^2}$ (B) $\frac{g}{\omega}$
 (C) $\frac{g\omega^2}{4\pi^2}$ (D) $\frac{g\omega}{2\pi}$

- 9 Which of the following statements is true?

- (A) $\frac{1}{b} > \frac{1}{a}$ if and only if $a > b$.
 (B) $\Phi^{-1}(a) > \Phi^{-1}(b)$ if and only if $a > b$, where $\Phi(x)$ is the standard cumulative normal distribution function.
 (C) $f'(a) > f'(b)$ if and only if $a > b$ where $f(x)$ is some differentiable function at $x = a$ and $x = b$.
 (D) $\int_a^b f(x) dx > \int_a^b g(x) dx$ if and only if $f(x) > g(x)$ for all $a \leq x \leq b$, where $f(x)$ and $g(x)$ are some continuous functions over all real x .

- 10 Which of the following inequalities is FALSE?

- (A) $\int_0^1 \sqrt{x} \cos^{-1} x dx < \frac{\pi}{2}$ (B) $\int_1^e \frac{\tan^{-1} x}{x} dx < \frac{\pi}{2}$
 (C) $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} x^4 \sin x dx < \frac{\pi}{15}$ (D) $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\cos x}{\sqrt{x}} dx < \frac{\pi}{15}$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet

(a) Evaluate

3

$$\int_0^1 \frac{x^2 \ln(1 + x + x^9 + x^{10})}{1 + x^3} dx.$$

(b) The coordinates of the centre of a circular plane in space is $(4, -1, 3)$.
Suppose that $(1, -1, 7)$ lies on the circular edge of the plane.

2

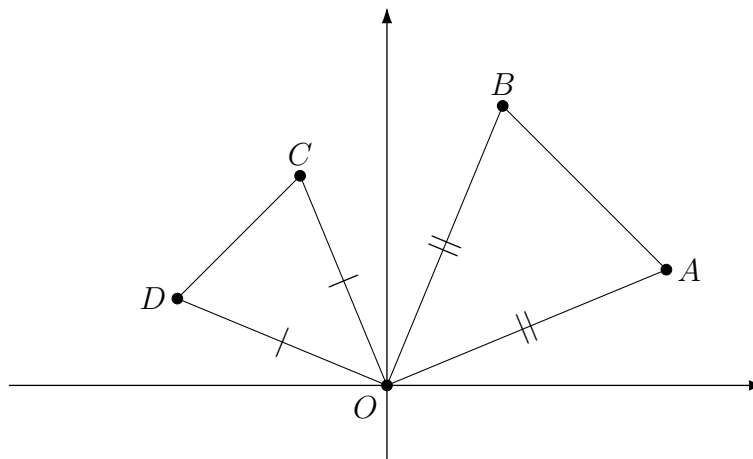
Find a parametric vector equation of the plane's circular edge. Express your answer in the form

$$f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

where t is a parameter, and $f(t)$, $g(t)$ and $h(t)$ are some functions of t .

(c) Let $\triangle OAB$ and $\triangle OCD$ be two similar but non-overlapping isosceles triangles on the complex plane with $OA = OB$ and $OC = OD$ as shown in the diagram below.

1



Let the complex number representing the points A, B, C and D be a, b, c and d respectively.

Show that $|a - c| = |b - d|$.

Question 11 continues on page 6

Question 11 (continued)

- (d) A line ℓ_1 has the vector equation $\underline{r} = 10\underline{i} + 8\underline{j} + 5\underline{k} + \lambda(\underline{i} + \underline{j} + \underline{k})$. Another line ℓ_2 intersects ℓ_1 at point M and passes through the points $A(4, 1, 3)$ and $B(6, 5, -3)$.

(i) Show that ℓ_1 and ℓ_2 are perpendicular. **1**

(ii) Let C and D be points on ℓ_1 such that M is the midpoint of CD and the distance from A to C is $\sqrt{62}$. Find the coordinates of C and D . **3**

- (e) Suppose that z is a complex number with modulus r and argument θ . Let R_1 be the region on the complex plane defined by the inequality

$$0 \leq \arg(z - 1) \leq \frac{\pi}{3}.$$

(i) Sketch the region R_1 . **1**

(ii) Suppose that $\omega = r(\sin \theta + i \cos \theta)$ where r and θ are specific values such that z lies in the region R_1 . **1**

Let R_2 be the region on the complex plane that represents the locus of ω . Sketch the region R_2 .

(iii) The region on the complex plane which represents the set of complex numbers which lie in both R_1 and R_2 can be described by the inequality **3**

$$\theta_1 \leq \arg(z - z_0) \leq \theta_2.$$

Find z_0, θ_1 and θ_2 .

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet

- (a) Let $P(z) = z^4 + 3z^2 + z - 1$. How many roots of $P(z)$ have a positive real part? **3**
Justify your answer.

- (b) Let $I_n = \lim_{a \rightarrow \infty} \left(\int_{-a}^a \frac{x^n e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} dx \right)$ for some non-negative integer n .

- (i) Show that $I_{n+2} = (n+1)I_n$. **2**

You may assume without proof that $\lim_{x \rightarrow \infty} \left(x^n e^{-\frac{x^2}{2}} \right) = 0$.

- (ii) Hence, show that $I_{2n} = \frac{(2n)!}{2^n n!}$. **1**

- (c) Consider the polynomial $P(z) = z^{2mn} - az^{mn} + 1$ for some positive integers m, n and real value $-2 < a < 2$. Suppose that $Q(z) = z^{2n} - az^n + 1$ is a factor of $P(z)$.

- (i) Show that the roots of both $P(z)$ and $Q(z)$ must satisfy $|z| = 1$. **2**

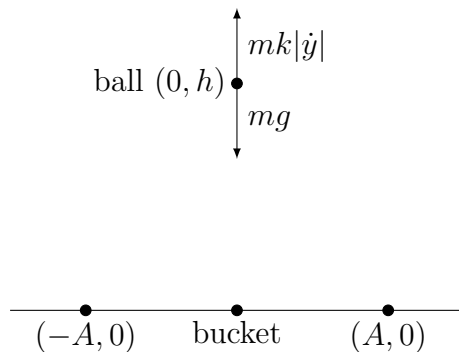
- (ii) Hence, find all possible values of a in terms of m and n . **2**

Question 12 continues on page 8

Question 12 (continued)

- (d) On the x - y plane, a ball of mass m is dropped vertically from rest at a sufficiently large height from the point $(0, h)$ above the origin. Whilst falling, the ball has a vertical displacement of y relative to the ground at time t . It is subjected to air resistance of magnitude $mk|\dot{y}|$, where k is a positive constant.

On the ground, a bucket is mounted on a machine which moves horizontally in simple harmonic motion with displacement x at time t relative to its centre of motion and amplitude A . The bucket is initially at its centre of motion $(0, 0)$.



- (i) By considering the relevant acceleration equation, show that

2

$$\dot{y} = \frac{g}{k}(e^{-kt} - 1)$$

where g is the acceleration due to gravity.

- (ii) The machine holding the bucket moves according to the acceleration equation $\ddot{x} = -\omega^2 x$ for some positive constant ω . Suppose that it is timed such that the bucket catches the ball at exactly the n th time it crosses its centre of motion (excluding its initial position).

3

Show that

$$\frac{\omega kh}{\pi g} < n < \frac{\omega}{\pi} \left(\frac{kh}{g} + \frac{1}{k} \right).$$

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet

- (a) Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_1 x + a_0$ where $a_0, a_1, a_2, \dots, a_n$ are integers with $a_0 \neq 0$ and $a_n \neq 0$. Suppose that $x = \frac{p}{q}$ is a root of $P(x)$ where p and q are integers with no common factors, other than 1.

(i) Show that a_0 is divisible by p and a_n is divisible by q . **2**

(ii) Hence, prove that $\cos \frac{\pi}{9}$ is irrational. **2**

You may use the result $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ without proof.

- (b) The points $P(p^\alpha, p^{1-\alpha})$ and $Q(q^\alpha, q^{1-\alpha})$ lie on the curve $y = x^{\frac{1-\alpha}{\alpha}}$ for some constants $p > 0$, $q > 0$ and $0 < \alpha < 1$.

(i) By considering areas, or otherwise, show that $p^\alpha q^{1-\alpha} \leq \alpha p + (1 - \alpha)q$. **2**

(ii) Hence, show that for positive values x, y and z **1**

$$\frac{x + y + z}{3} \geq (xyz)^{\frac{1}{3}}.$$

(iii) Let x_i be a positive real number for $i = 1, 2, \dots, n$. Let α_i be a positive real number such that $\alpha_1 + \alpha_2 + \cdots + \alpha_n = 1$ and $0 < \alpha_i < 1$ for $i = 1, 2, \dots, n$. Use the result in part (i) to prove by mathematical induction that for integer $n \geq 2$ **3**

$$\alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_n x_n \geq x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}.$$

(iv) Let X be a discrete random variable which takes the values $x_1, x_2, \dots, x_n$ with probabilities $\alpha_1, \alpha_2, \dots, \alpha_n$ respectively. Use the result in part (iii) to show that **1**

$$\text{Var}(\sqrt{X}) \leq \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_n x_n - x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}.$$

- (c) A particle moves in simple harmonic motion along the x -axis with its centre of motion at $x = 1$. The square of the particle's velocity v^2 at time t has the relation

$$v^2 = 4 - 4 \sin 4t.$$

(i) Find the time that the particle is first at rest. **1**

(ii) Hence, find the displacement equations of the particle in terms of t . **3**

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet

- (a) (i) Express $\frac{5}{(x-2)(2x+1)}$ as a sum of partial fractions. **1**

- (ii) Use the substitution $t = \tan \frac{x}{2}$ to show that for some constant c_1 **3**

$$\int \frac{5}{4 \cos x + 3 \sin x} dx = \ln \left(\frac{\sin x - 2 \cos x + 2}{2 \sin x + \cos x - 1} \right) + c_1$$

where $0 \leq x \leq \frac{\pi}{2}$.

- (iii) By writing $4 \cos x + 3 \sin x$ in the form $R \cos(x - \alpha)$ for some real numbers R and α , show that for some constant c_2 **3**

$$\int \frac{5}{4 \cos x + 3 \sin x} dx = \ln \left(\frac{4 \sin x - 3 \cos x + 5}{3 \sin x + 4 \cos x} \right) + c_2$$

where $0 \leq x \leq \frac{\pi}{2}$.

- (iv) Hence, carefully explain why for $0 \leq x \leq \frac{\pi}{2}$ **1**

$$\frac{\sin x - 2 \cos x + 2}{2 \sin x + \cos x - 1} \equiv \frac{4 \sin x - 3 \cos x + 5}{3 \sin x + 4 \cos x}.$$

- (b) Let $P(x) = x^{2n} + x^{2n-1} + x^{2n-2} + \dots + x + 1$.

- (i) Find all complex roots of $P(x)$. **1**

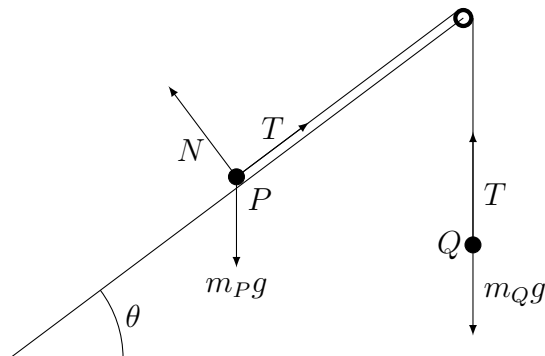
- (ii) Hence, show that **2**

$$\tan \left(\frac{\pi}{2n+1} \right) \tan \left(\frac{2\pi}{2n+1} \right) \tan \left(\frac{3\pi}{2n+1} \right) \dots \tan \left(\frac{2n\pi}{2n+1} \right) = (-1)^n (2n+1).$$

Question 14 continues on page 11

Question 14 (continued)

- (c) An object P of mass m_P sits on a smooth inclined slope at a fixed acute angle of θ to the horizontal. It is held by a light string which extends up to the edge of the slope into a smooth pulley as shown in the diagram below. Another object Q of mass m_Q is held vertically by the same light string from that pulley over the edge of the inclined slope.



The forces acting on object P are gravity (with acceleration g), the tension on the string of magnitude T and a normal force of magnitude N perpendicular to the slope. The forces acting on object Q are gravity and tension on the string T .

The net forces on each object are such that they each maintain a constant speed at any point in time.

- (i) By resolving the forces on both objects, show that $m_P > m_Q$. 1
- (ii) Suppose that the inclined slope is rough instead and object P is subjected to a friction force F opposing its direction of motion. As long as the ratio $F : N$ is no more than $1 : \sqrt{3}$ then neither object will accelerate in any direction. 3

Show that if the ratio of $m_P : m_Q$ is $\sqrt{3} : \sqrt{2}$ and neither object accelerates in any direction, then $\frac{\pi}{12} \leq \theta \leq \frac{5\pi}{12}$.

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet

- (a) Suppose that $\triangle ABC$ is an acute angled triangle.

4

Let P lie on AB such that the ratio $|\overrightarrow{AP}| : |\overrightarrow{PB}|$ is 1 : 2 and let Q lie on AC such that the ratio $|\overrightarrow{AQ}| : |\overrightarrow{QC}|$ is 1 : 3.

Let O be intersection of the intervals BQ and CP . The interval AO is produced until it intersects with BC at R .

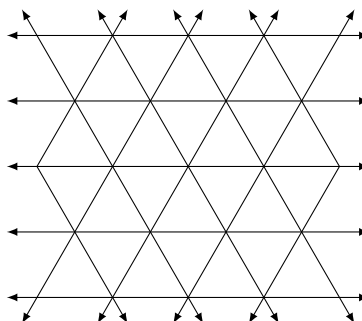
By considering the relevant vector additions, find the ratio of $|\overrightarrow{BR}| : |\overrightarrow{RC}|$.

- (b) Suppose that the x - y plane is made up of a set of individual points. Each point in the plane is assigned a colour. Suppose that the assignment is such that any two points 1 unit apart on the plane will *always* have different colours. There are N distinct colours in total.

- (i) A triangular lattice on a plane is provided below where all triangles are congruent and equilateral with each side having a length of 1 unit.

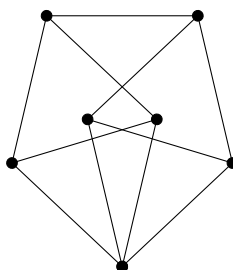
1

By considering the intersection points in this triangular lattice, explain why $N \geq 3$.



- (ii) Now consider seven points on the plane arranged as shown below. Each interval drawn between two points in the diagram has unit length.

2



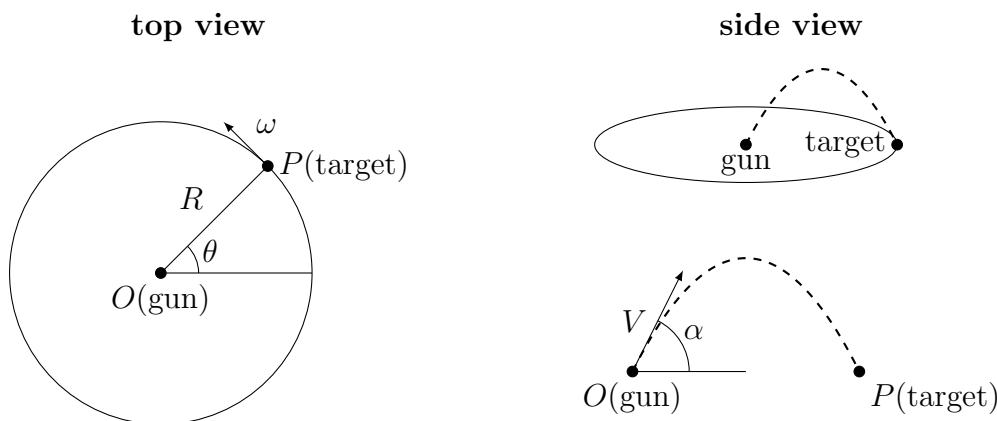
By considering the colour of each of these points, prove that $N \geq 4$.

Question 15 continues on page 13

Question 15 (continued)

- (c) An automated gun is fixed at position O on the ground pointing at an angle of elevation of α to the horizontal. A target P circles the gun in a horizontal plane on the ground with centre O and radius R .

The gun fires a bullet from O in the air at an initial speed of V . At time t seconds after the gun was fired, the target P has displaced an angle of θ at the centre O , where $\theta > 0$. The target P moves along the circle such that the rate of change of θ is a constant positive value ω . Assume that the gun is orientated such that it will always aim for the target after firing.



Let g be the acceleration due to gravity. The displacement vector of the bullet is

$$\vec{r} = Vt \cos \alpha \vec{i} + \left(Vt \sin \alpha - \frac{gt^2}{2} \right) \vec{j}. \quad (\text{Do NOT prove this})$$

- (i) Show that if $V < \sqrt{gR}$, the bullet will never hit the target P . 1
- (ii) Suppose that $V \geq \sqrt{gR}$ and the bullet hits the target P . Show that the corresponding value(s) of θ must satisfy 3

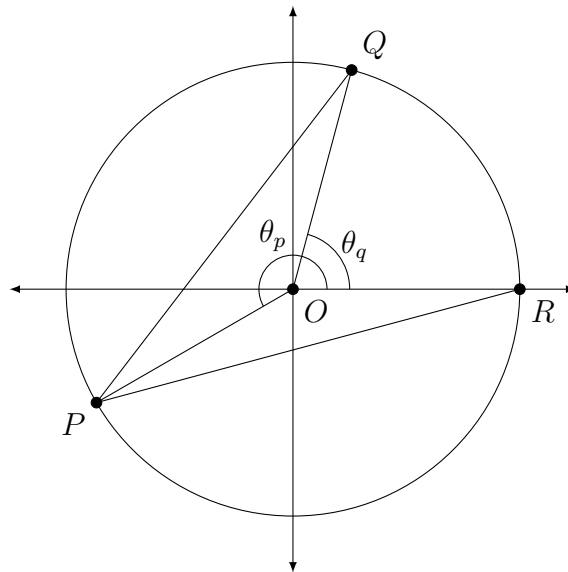
$$g^2\theta^4 - 4\omega^2V^2\theta^2 + 4R^2\omega^4 = 0.$$
- (iii) Explain why if $V > \sqrt{gR}$ then there are two distinct values of θ where the bullet hits the target. 1
- (iv) Let θ_1 and θ_2 be the distinct values of θ corresponding to the points P_1 and P_2 respectively, where the bullet hits the target. 3

Show that the length of the interval P_1P_2 is $2R \left| \sin \left(\frac{\omega \sqrt{V^2 - gR}}{g} \right) \right|$.

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet

- (a) On the complex plane, let P and Q be points on a circle of radius r with the centre as the origin, as shown on the diagram below. Let R be the point on the circle that intersects the positive real axis.



Let the complex numbers representing P and Q be p and q respectively.

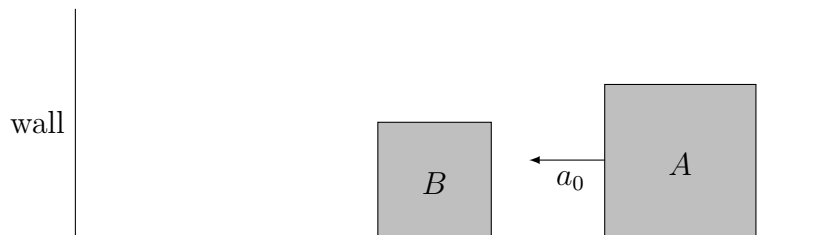
Suppose that $\arg p = \theta_p$ and $\arg q = \theta_q$ where $\theta_p \in [0, 2\pi)$, $\theta_q \in [0, 2\pi)$ and $\theta_p > \theta_q$.

- (i) Show that $\angle QPR = \arg \left(\frac{e^{i\theta_q} - e^{i\theta_p}}{1 - e^{i\theta_p}} \right)$. **1**
- (ii) Hence, prove that $\angle QOR = 2\angle QPR$. **2**

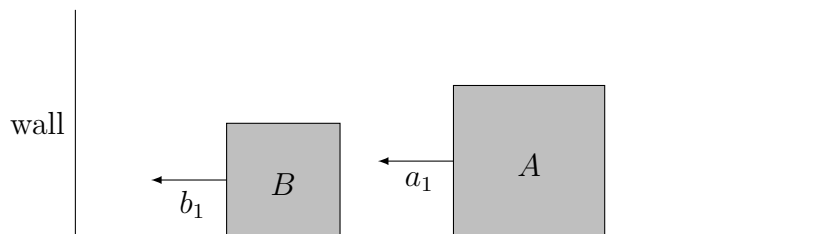
Question 16 continues on page 15

Question 16 (continued)

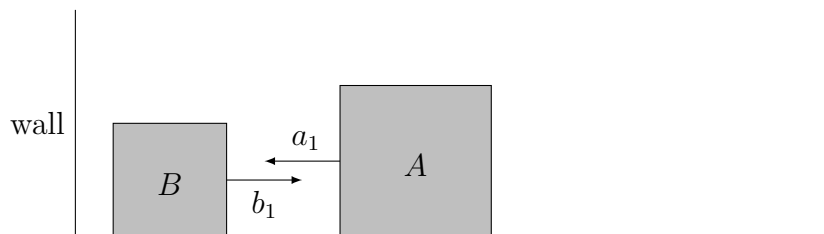
- (b) An object A of mass m_A moves in a horizontal line on a friction-less surface at a constant speed a_0 towards a second object B of mass m_B , initially at rest.



Object A collides with object B , which subsequently moves object B towards a stationary wall with speed b_1 and object A decelerates to a speed of a_1 in the same direction.



Object B then collides with the wall and moves back towards object A at the same speed b_1 but opposite direction, before colliding with it. The wall remains stationary at all times.



This process repeats until both objects are moving away from the wall and the speed of object A exceeds the speed of object B so they do not collide again.

Let v_A and F_A be the velocity and force of object A respectively at time t , where the positive direction is to the right. Let v_B and F_B be similarly defined.

Question 16 continues on page 16

Question 16 (continued)

- (i) Explain why during the collision between object A and object B that 1

$$F_A + F_B = 0.$$

- (ii) Let $p = v_A\sqrt{m_A}$ and $q = v_B\sqrt{m_B}$. By solving the differential equations associated with the force equation in part (i), show that at the first collision 3

$$p^2 + q^2 = a_0^2 m_A \quad \text{and} \quad p\sqrt{m_A} + q\sqrt{m_B} = -a_0 m_A.$$

- (iii) Hence, on the p - q axes (with p being the horizontal axis), plot the two points corresponding to the velocities of both objects just before and just after the first collision with each other. Justify your answer. 2

- (iv) On the same diagram using the p - q axes, plot the next two points corresponding to the velocities of both objects just before and just after their second collision with each other. Justify your answer. You do not need to identify the exact coordinates of the two points. 2

- (v) Let N be the total number of collisions that occur before both objects move away from the wall and never collide again. This includes collisions between object A and object B as well as collisions between object B and the wall. 4

Using the result in part (a)(ii), show that for $N > 2$

$$\tan^2\left(\frac{\pi}{N+1}\right) < \frac{m_B}{m_A} < \tan^2\left(\frac{\pi}{N}\right).$$

End of paper